down-sampling. **c**, Schematic illustration of 2 cell selection strategies: left, spatial-distance-based selection of the 5 nearest neighbors (used in CELLECT); right, feature-distance-based selection of the 5 most similar feature vectors. **d**, Error curves (FN + FND + IS) over time for both strategies under different temporal down-sampling. The feature distance-based strategy performs comparably at early frames when cell density is low, even under reduced temporal resolution, but its performance degrades rapidly as cell density increases over time. **e**, Circular tree diagrams displaying tracking errors across all 6 settings. All evaluation data were drawn from mskcc-confocal dataset #2; the model was trained on mskcc-confocal dataset #3 without temporal down-sampling. Scale bar: 10 μm.